



Simplified Instructional Computer(SIC) Machine Architecture



SIC Machine Architecture

- The SIC machine architecture depends on the following features:
 - Memory
 - Registers
 - Data Formats
 - Instruction Formats
 - Addressing Modes
 - Instruction Set
 - Input and Output

SIC Machine Architecture

- Memory
 - Memory consists of 8-bit bytes
 - Any 3 consecutive bytes form a word (24 bits)
 - Total of 32768 (2^{15}) bytes in the computer memory



SIC Machine Architecture

- Registers
 - Five 24-bits registers. Their mnemonic, number and use are given in the following table.

Mnemonic	Number	Use
A	0	Accumulator; used for arithmetic operations
X	1	Index register; used for addressing
L	2	Linkage register; contains the return address whenever control transferred to subroutine
PC	8	Program counter; contains the address of the next instruction to be executed.
SW	9	Status word, including Condition code such as $<, \leq, >, \geq, ==$.

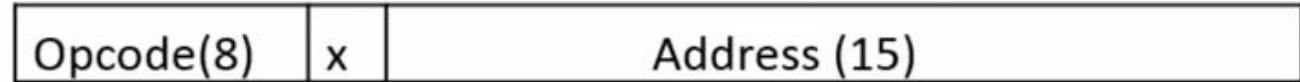
SIC Machine Architecture

- Data Formats
 - Integers are stored as 24-bit binary number
 - 2's complement representation for negative values
 - Characters are stored using 8-bit ASCII codes
 - No floating-point hardware on the standard version of SIC

SIC Machine Architecture

- Instruction Formats

- All machine instructions on the standard version of SIC have the 24-bit format as shown below



- Addressing Modes

- There are two addressing modes available, which are as shown in the below table. Parentheses are used to indicate the contents of a register or a memory location.

Mode	Indication	Target address calculation
Direct	$x = 0$	$TA = \text{address}$
Indexed	$x = 1$	$TA = \text{address} + (x)$

SIC Machine Architecture

- Instruction Set
 - Load and store registers - LDA, LDX, STA, STX, etc.
 - Integer arithmetic operations - ADD, SUB, MUL, DIV
 - All arithmetic operations involve register A and a word in memory, with the result being left in A
 - COMP – Comparison instruction
 - Conditional jump instructions - JLT, JEQ, JGT
 - Subroutine linkage - JSUB, RSUB
 - I/O (transferring 1 byte at a time to/from the rightmost 8 bits of register A)
 - Test Device instruction (TD)
 - Read Data (RD)
 - Write Data (WD)

SIC/XE Machine Architecture

- The SIC/XE machine architecture depends on the following features:
 - Memory
 - Registers
 - Data Formats
 - Instruction Formats
 - Addressing Modes
 - Instruction Set
 - Input and Output

SIC/XE Machine Architecture

- Memory
 - Memory consists of 8-bit bytes
 - Any 3 consecutive bytes form a word (24 bits)
 - Total of 1 Mb (2^{20}) bytes in the computer memory

SIC/XE Machine Architecture



- Registers

- There are nine registers; each register is 24 bits in length except floating point register.
- Their mnemonic, number and uses are shown in the following table.

Mnemonic	Number	Use
A	0	Accumulator; used for arithmetic operations
X	1	Index register; used for addressing
L	2	Linkage register; contains the return address whenever control transferred to subroutine
B	3	Base register; used for addressing
S	4	General working register-no special use
T	5	General working register-no special use
F	6	Floating point accumulator (48 bits)
PC	8	Program counter; contains the address of the next instruction to be executed.
SW	9	Status word, including Condition code such as <, ≤, >, ≥, =, ≠

SIC/XE Machine Architecture

- Data Formats
 - Integers are stored as 24-bit binary number
 - 2's complement representation for negative values
 - Characters are stored using 8-bit ASCII codes
 - Support 48 bit floating-point numbers

1	11	36
s	exponent	fraction

- There is a 48-bit floating-point data type, $F \cdot 2^{(e-1024)}$

SIC/XE Machine Architecture

- Instruction Formats

- Format 1 (1 byte) –

8
opcode

 Example: RSUB

- Format 1 (2 byte) -

8	4	4
op	r1	r2

 Example: ADDR S, T

- Format 3 (3 byte) -

6	1	1	1	1	1	1	12
op	n	i	x	b	p	e	Displacement

 - Example: LDA #3

- Format 4 (4 byte) -

6	1	1	1	1	1	1	20
op	n	i	x	b	p	e	Address

 - Example: +JSUB RDREC

SIC/XE Machine Architecture

- Addressing Modes and Flag Bits
 - Base relative ($n=1, i=1, b=1, p=0$)
 - Program-counter relative ($n=1, i=1, b=0, p=1$)
 - Direct ($n=1, i=1, b=0, p=0$)
 - Immediate ($n=0, i=1, x=0$)
 - Indirect ($n=1, i=0, x=0$)
 - Indexing (both n & $i = 0$ or $1, x=1$)
 - Extended ($e=1$ for format 4, $e=0$ for format 3)

SIC/XE Machine Architecture

- Instruction Set
 - Load and store registers - LDA, LDX, STA, STX, LDB, STB etc.
 - Integer arithmetic operations - ADD, SUB, MUL, DIV
 - Floating-point arithmetic operations: ADDF, SUBF, MULF, DIVF
 - COMP – Comparison instruction
 - Conditional jump instructions - JLT, JEQ, JGT
 - Subroutine linkage - JSUB, RSUB
 - Register move instruction: RMO
 - Register-to-register arithmetic operations: ADDR, SUBR, MULR, DIVR

SIC/XE Machine Architecture

- I/O (transferring 1 byte at a time to/from the rightmost 8 bits of register A)
 - Test Device instruction (TD)
 - Read Data (RD)
 - Write Data (WD)